

INDIVIDUAL PROJECT SUMMARY REPORT

Calibo AI Academy – Phase 1 Mini Use Case 01

Student Name	HARITHA VELAGAPUDI
Project Title	Category Intelligence – Retail Product Performance EDA
Individual Role	Data Analyst
Tools Used	Python, Pandas, Matplotlib, Jupyter Notebook
Dataset	Retail Sales Dataset

CBIM Problem Canvas

Business Problem: Retail sales data contains category-level and time-based information, but raw data does not directly show how category revenue changes over time.

Business Objective: Analyze the retail dataset to understand category revenue performance and convert the results into clear business insights that can support category management and decision-making.

Data Used: Retail Sales Dataset containing sales information used for category and time-based analysis.

Key Analysis Focus: Step 1 – Load & Inspect the dataset; Step 2 – Category Revenue Trend.

Expected Business Output: A clean understanding of the dataset structure and a category-wise monthly revenue trend that highlights changes in performance.

1. Project Overview

The project focuses on analyzing retail sales data from a Data Analyst perspective. The main objective is to transform raw sales data into structured information and business-oriented insights. The analysis begins with loading and inspecting the dataset and then moves to category-wise revenue trend analysis.

2. My Individual Contribution – Data Analyst

As the Data Analyst, I worked on the initial data understanding and category revenue analysis. In Step 1, I loaded the retail dataset using Python and Pandas and inspected its structure, columns, data types, and records. I checked the dataset for missing values and duplicate records and converted the date column into a proper date-time format where required.

In Step 2, I analyzed the category revenue trend. I extracted the month from the date, grouped the data by month and category, and calculated monthly total revenue. I presented the results in a table and visualized the monthly revenue trends using Matplotlib. I then examined the highest and lowest revenue periods, compared category performance across months, and identified important changes in the trend for business interpretation.

3. Step 1 – Load & Inspect

Purpose: Understand the dataset before performing analysis. The dataset was loaded with Pandas and inspected to identify its structure, columns, data types, number of records, missing values, and duplicate records.

Data Analyst Output: The inspection provided the basic data-quality and structural understanding required before category revenue calculations. The date field was prepared for time-based analysis.

4. Step 2 – Category Revenue Trend

Purpose: Determine how revenue for each retail category changes month by month.

Method: Convert the date into a date-time format, create a month field, group the data by month and category, aggregate revenue, and visualize the resulting trends.

Categories analyzed: Electronics, Apparel, Grocery, Home & Kitchen, and Personal Care.

Business Interpretation: The category revenue trend helps identify periods of stronger or weaker category performance, compare categories across the six-month period, and provide evidence for further category-level business decisions.

5. Skills and Concepts Applied

The main concepts applied were data loading, data inspection, data-quality checking, date-time conversion, grouping, aggregation, sorting, monthly trend analysis, and data visualization. Pandas was used for data manipulation and aggregation, while Matplotlib was used to represent category revenue trends visually.

6. Learning Outcomes

Through this role, I learned how a Data Analyst approaches a business dataset from the initial inspection stage to meaningful trend analysis. I improved my understanding of data preparation, time-based grouping, category-level revenue analysis, visualization, and communicating analytical findings in a simple business-oriented manner.

7. Conclusion

My contribution as a Data Analyst covered the foundation of the Category Intelligence use case: understanding the raw retail dataset and analyzing category revenue trends over time. Step 1 established data readiness, while Step 2 converted the prepared data into a clear category-wise monthly revenue view. Together, these steps support the team's broader Category Intelligence analysis and business briefing.